IF 451
Advance Web Programming

Meet3 - JSON Basics & CRUD with Files

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Meet 3

Objective:
Sub-CLO0712
Students are able to build basic web programming and feature in Node JS (C6)

OUTLINE

- JSON structure.
- Reading & writing JSON files.
- *Adding, updating, deleting entries.*

JSON structure

JSON (JavaScript Object Notation)

JSON is a lightweight, text-based (string) data format used for exchanging data between applications.

It's similar to JavaScript objects, but more restrictive (for example, properties must be enclosed in quotation marks " ").

JSON Structure = data in key-value format similar to an object.

Can be used to exchange data between servers, applications, or APIs.

In Node.js:

- JSON.stringify() → converts an object → JSON string.
- JSON.parse() → converts a JSON string → object.
- Can be saved/retrieved from a .json file.

Sample JSON


```
[
  {
    "nama": "Wirawan",
    "hobi": ["membaca", "coding", "musik"],
    "alamat": {
      "kota": "Tangerang",
      "kodepos": 12345
    }
  },
  {
    "nama": "Ika",
    "hobi": ["main games", "berenang"],
    "alamat": {
      "kota": "Jakarta",
      "kodepos": 54321
    }
  }
]
```

JSON Structure

Main JSON Object

Array of Objects

Array of Objects

```
[
  {
    "nama":“Wirawan",
    "hobi": ["membaca", "coding", "musik"],
    "alamat": {
      "kota":“Tangerang",
      "kodepos": 12345
    }
  },
  {
    "nama":“Ika",
    "hobi": [“main games",“berenang"],
    "alamat": {
      "kota":“Jakarta",
      "kodepos": 54321
    }
  }
]
```

Element data with single value

Element data with mutliple value in array

Element data with sub-attribute and value

Create JSON data in NodeJS

Create a JavaScript Object and then convert it to JSON (use JSON.stringify)

1. Create new file JS
2. Write the script beside,
3. Run with **nodemon youfilename.js**

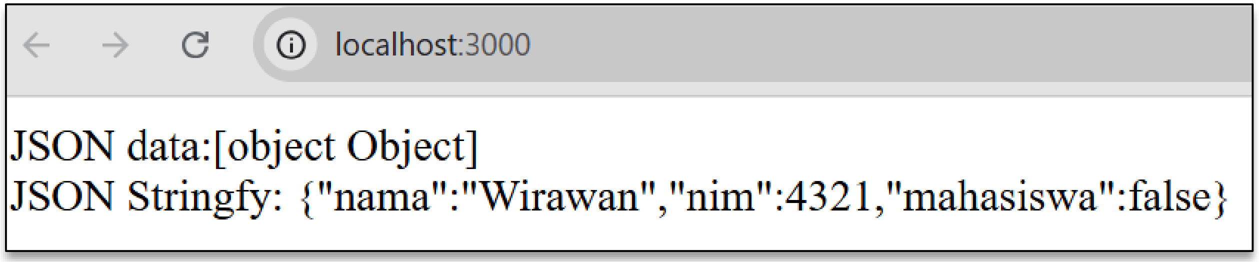
```
const data = {
      nama: "Wirawan",
      nim: 4321,
      mahasiswa: false
  };

console.log("JSON  data:" + data); // object biasa
console.log("JSON Stringfy: " + JSON.stringify(data)); // ubah ke JSON string
```

```
JSON  data:[object Object]
JSON Stringfy: {"nama":"Wirawan","nim":4321,"mahasiswa":false}
```

Show JSON result into Website

```
const http = require("http");

const server = http.createServer((req, res) => {
    res.writeHead(200, {"Content-Type":"text/html"});

    const data = {
        nama: "Wirawan",
        nim: 4321,
        mahasiswa: false
    };

    res.write("JSON  data:" + data);
    res.write("<br>JSON Stringfy: " + JSON.stringify(data));

    res.end();
});
```

```
server.listen(3000, ()=>{
    console.log("server berjalan di
http://localhost:3000");
});
```

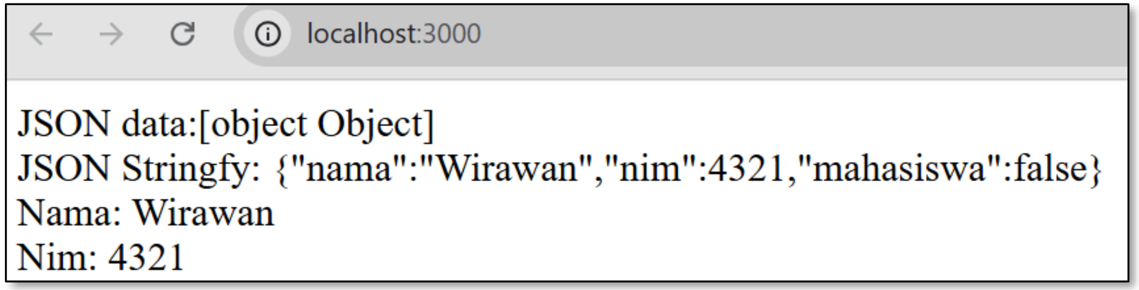


Reading & writing JSON files.

Read JSON String then parse to Object (use JSON.parse) – **R**ead

1. From the previous code, edit or add the code like the sample beside to get single data information.
2. Refresh your browser and see the result

```
const server = http.createServer((req, res) => {
    res.writeHead(200, {"Content-Type":"text/html"});

    const data = {
        nama: "Wirawan",
        nim: 4321,
        mahasiswa: false
    };

    res.write("JSON  data:" + data);
    res.write("<br>JSON Stringfy: " + JSON.stringify(data));

    const obj = JSON.parse(JSON.stringify(data));
    res.write("<br>Nama: " + obj.nama);
    res.write("<br>Nim: " + obj.nim);

    res.end();
});
```

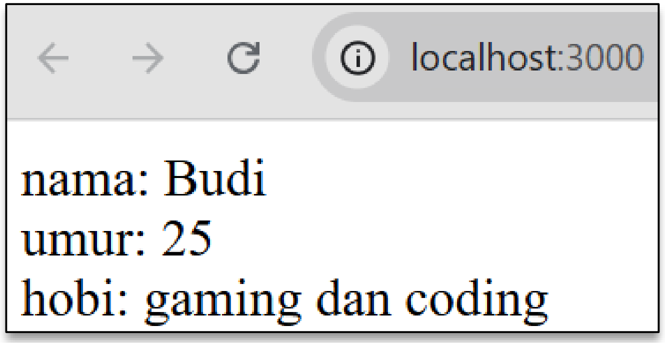


Read external JSON (fs = filesystem)

1. Create new file and save it with name "***data.json***"
2. In "data.json", write the code beside.
3. Back to your main **app.js** file, and write this code, to call external json file and show in your browser

```json
{
  "nama": "Budi",
  "umur": 25,
  "hobi": ["gaming", "coding"]
}
```

```javascript
const http = require("http");
const fs = require("fs");

const server = http.createServer((req, res) => {
    const rawData = fs.readFileSync("data.json");
    const data = JSON.parse(rawData);

    res.writeHead(200, {"Content-Type":"text/html"});
    res.write("nama: " + data.nama);
    res.write("<br>umur: " + data.umur);
    res.write("<br>hobi: " + data.hobi[0] + " dan " + data.hobi[1]);

    res.end();
});

server.listen(3000, ()=>{
    console.log("server berjalan di http://localhost:3000");
});
```



Multiple Object JSON

See JSON sample beside, how to read and show it all data in NodeJS?

1. Crete new json file, with name "***data2.json***"
2. Write the json code beside.

```json
[
  {
    "nama": "Indah",
    "umur": 19,
    "mahasiswa": true,
    "hobi": ["menari", "belajar"],
    "alamat": {
      "kota": "Jogja",
      "kodepos": 13579
    }
  },
  {
    "nama": "Toni",
    "umur": 21,
    "mahasiswa": true,
    "hobi": ["membaca", "coding game", "tidur"],
    "alamat": {
      "kota": "Bandung",
      "kodepos": 54321
    }
  }
]
```

Show Multiple Object JSON (single)

In your main **app.js,** write the code beside

Conclusion:

1. When you want to call your first array object, using [number]
2. If the attribut have multiple sub-attribut, call it, with dot (.)

```
const http = require("http");
const fs = require("fs");

const server = http.createServer((req, res) => {
    const rawData = fs.readFileSync("data2.json");
    const data = JSON.parse(rawData);

    res.writeHead(200, {"Content-Type":"text/html"});
    res.write("nama: " + data[0].nama);
    res.write("<br>umur: " + data[0].umur);
    res.write("<br>Hobi: " + data[0].hobi[0] + " dan " + data[0].hobi[1]);
    res.write("<br>Alamat: " + data[0].alamat.kota);
    res.write("<br>Kodepos: " + data[0].alamat.kodepos);

    res.end();
});

server.listen(3000, ()=>{
    console.log("server berjalan di http://localhost:3000");
});
```

Show Multiple Object JSON (multiple)

In your main **app.js,** write the code beside

Conclusion:

1. Use **dataJSONParse.foreach** to call all object
2. You can change "element" with other keyword

```javascript
const http = require("http");
const fs = require("fs");

const server = http.createServer((req, res) => {
    const rawData = fs.readFileSync("data2.json");
    const data = JSON.parse(rawData);

    res.writeHead(200, {"Content-Type":"text/html"});

    data.forEach(element => {
        res.write("nama: " + element.nama);
        res.write("<br>umur: " + element.umur);
        res.write("<br>Hobi: " + element.hobi.join(", "));
        res.write("<br>Alamat: " + element.alamat.kota);
        res.write("<br>Kodepos: " + element.alamat.kodepos);
        res.write("<hr>");
    });

    res.end();
});

server.listen(3000, ()=>{
    console.log("server berjalan di http://localhost:3000");
});
```

Adding, updating, deleting entries.

Adding Data JSON - **C**reate

1. To add data to JSON file, please create new file “***app_add_json.js***”
2. Make sure, your json file, have [ ] to convert single object to array object
3. Write the code beside into your “***app_add_json.js***” files
4. And run your files, from cmd, with command “node *app_add_json.js*”

```json
[
  {
    "nama": "Budi",
    "umur": 25,
    "hobi": [
      "gaming",
      "coding"
    ]
  }
]
```

```javascript
const fs = require("fs");

let rawData = fs.readFileSync("data.json");
let data = JSON.parse(rawData);

// Jika data bukan array, bungkus dulu jadi array
if (!Array.isArray(data)) {
  data = [data];
}

// Cek apakah sudah ada "Siti"
const exists = data.some(item => item.nama === "Siti");

if (!exists) {
    data.push({
        nama: "Siti",
        umur: 22,
        hobi: ["membaca", "menulis"]
    });

    fs.writeFileSync("data.json", JSON.stringify(data, null, 2));
    console.log("Data berhasil ditambahkan");
} else {
    console.log("Data sudah ada");
}
```

Edit Data JSON - **U**pdate

1. To edit data, create new file, with file name "***app_update_json.js***"
2. *Write the code beside to update data "siti"*
3. Run your files, from cmd, with command "node *app_update_json.js*

```
const fs = require("fs");

let rawData = fs.readFileSync("data.json");
let data = JSON.parse(rawData);

// [OPTIONAL] Pastikan data array
if (!Array.isArray(data)) {
  data = [data];
}

const index = data.findIndex(item => item.nama === "Siti");

if (index !== -1) {
  // Update field umur
  data[index].umur = 123;

  // Tambah hobi baru
  data[index].hobi.push("berenang");

  fs.writeFileSync("data.json", JSON.stringify(data, null, 2));
  console.log("Data Siti berhasil diupdate!");
} else {
  console.log("Data Siti tidak ditemukan.");
}
```

Delete Data JSON - **D**elete

1. To delete data, create new file, with file name "***app_delete_json.js***"
2. *Write the code beside to delete data "siti"*
3. Run your files, from cmd, with command "node *app_delete_json.js*

```
const fs = require("fs");

let rawData = fs.readFileSync("data.json");
let data = JSON.parse(rawData);

// [OPTIONAL] Pastikan data berupa array
if (!Array.isArray(data)) {
  data = [data];
}

// Hapus data dengan filter
data = data.filter(item => item.nama !== "Siti");
fs.writeFileSync("data.json", JSON.stringify(data, null, 2));

console.log("Data Siti berhasil dihapus!");
```

NEXT WEEK'S OUTLINE

- Mini Project: Student Management (JSON)

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Visi

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2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.